confidence about what the resulting state was. This means the agent has a high degree of control over the environment, since it can reliably reach states given its actions. Hence, a maximally empowered agent has maximal degrees of freedom.

This is different than the prediction-error approach because minimizing PE directly encourages exploration, whereas maximizing empowerment may induce exploratory behavior as a means to learn empowering skills, but only indirectly. Consider a young woman, Sarah, who decides to travel the world and explore as much as possible. She is reducing her uncertainty about the world. Compare her to Bill Gates, who by being extraordinarily rich, has a high degree of power. He may not be interested in traveling as much as Sarah, but he can if he wants, and no matter where he is at any time, he can go where he wants to go.

Both empowerment and curiosity objectives have their use cases. Empowerment-based objectives have been shown to be useful for training agents to acquire complex skills without any extrinsic reward (e.g., robotic tasks or sports games), whereas curiosity-based objectives tend to be more useful for exploration (e.g., games like Super Mario Bros. where the goal is to progress through levels). In any case, these two metrics are more similar than they are different.

Summary

- The sparse reward problem is when an environment rarely produces a useful reward signal, which severely challenges the way ordinary DRL attempts to learn.
- The sparse reward problem can be solved by creating synthetic reward signals that we call curiosity rewards.
- A curiosity module creates synthetic rewards based on how unpredictable the next state of the environment is, encouraging the agent to explore more unpredictable parts of the environment.
- The intrinsic curiosity module (ICM) consists of three independent neural networks: a forward-prediction model, an inverse model, and an encoder.